

Smart Public Facility Locator System

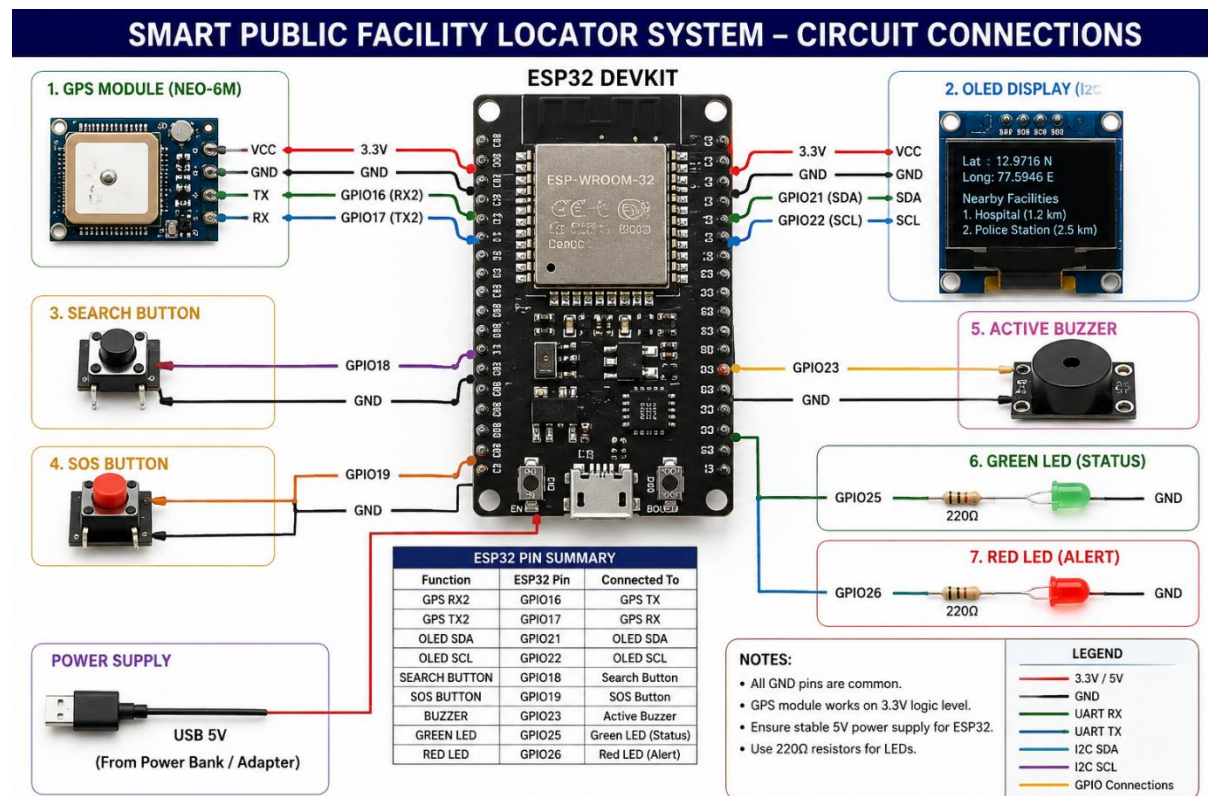
Abstract

The Smart Public Facility Locator System is a city solution that helps people find nearby public facilities. These include hospitals, police stations, fire stations, public toilets, pharmacies, bus stops, ATMs, parking areas and EV charging stations. The system uses GPS technology to find where the user is now. It also uses maps to show nearby facilities how far they are and the best way to get there. A cloud-based database keeps all the facility information. Let's people in charge update the details as needed. The system has features like help in emergencies searching for facilities giving feedback and getting directions. This makes it easier for people to find what they need. By making it faster to find public services the project helps with emergencies and better use of city resources. This solution helps build cities by offering reliable easy to use and up, to date location services. These services make the city safer easier to move in and better to live in.

Components Required :

S.No	Component	Quantity	Purpose
1	ESP32 Development Board	1	Main controller
2	GPS Module (NEO-6M)	1	Get current location
3	0.96" OLED Display (I2C)	1	Display location and facility details
4	Push Button	2	Search and SOS functions
5	Active Buzzer	1	Emergency alert indication
6	LED (Red)	1	Emergency status indicator
7	LED (Green)	1	System status indicator
8	220Ω Resistors	2	Current limiting for LEDs
9	Breadboard	1	Circuit assembly
10	Jumper Wires	As required	Connections
11	USB Cable	1	Programming and power
12	5V Power Supply / Power Bank	1	Power source

Circuit Diagram :



Introduction

A Smart Public Facility Locator System (SPFLS) is a smart city application that helps citizens quickly locate nearby public facilities such as

- Hospitals
- Ambulance services
- Police stations
- Fire stations
- Public toilets
- Bus stops
- Railway stations
- Parking areas
- Medical shops

Problem Statement :

- Unable to find nearby emergency services
- Difficulty locating public toilets
- Time wasted searching for hospitals
- Tourists don't know available facilities
- Parking availability is unknown
- No centralized public facility information

Features of Smart Public Facility Locator System :

1. GPS-Based Location Detection

The system automatically detects the user's current location using GPS technology. It provides accurate latitude and longitude coordinates for real-time tracking. This helps users find nearby public facilities quickly.

2. Public Facility Search

Users can search for facilities such as hospitals, police stations, ATMs, public toilets, pharmacies, and parking areas. The search feature is simple and fast. It helps users locate essential services with ease.

3. Interactive Digital Map

The system displays all nearby public facilities on an interactive map. Users can view the exact location of each facility. The map also allows users to select their preferred destination.

4. Route Navigation

The application provides the shortest and most efficient route to the selected facility. It also displays the estimated travel distance and travel time. This helps users reach their destination quickly.

5. Emergency Assistance

The system includes an Emergency SOS feature for urgent situations. Users can quickly locate nearby hospitals, police stations, or fire stations. This improves public safety and emergency response.

6. Real-Time Database Updates

Facility information is updated regularly through a cloud-based database. Users always receive the latest information about available public services. This increases the reliability of the system.

7. User Feedback and Ratings

Users can rate public facilities and submit feedback based on their experience. These reviews help improve service quality. Other users can also make better decisions using the ratings.

8. Admin Management

The administrator can add, edit, update, or remove facility information. This ensures that the database remains accurate and up to date. It also helps manage the entire system efficiently.

9. Secure User Authentication

The system provides secure user registration and login features. User information is protected through authentication mechanisms. This improves the security and privacy of the application.

10. Smart City Integration

The project supports smart city development by providing intelligent location-based public services. It improves accessibility to essential facilities for citizens and visitors. The system enhances urban management and public convenience.